HAZELCAST

Product Marketing Brief

Microservices Framework

A complete reference architecture that shows customers *how* to build microservices on Hazelcast — not just *that* they can.

100K+	<1ms	10	10	13	7
VIEW OPS/SEC	P99 LATENCY	MODULES	MCP AI TOOLS	BLOG POSTS	DASHBOARDS

Customers know Hazelcast is fast. What they often lack is a blueprint for putting that speed to work inside a microservices architecture. This framework fills that gap: a **four-service eCommerce application** built entirely on Hazelcast and Spring Boot, demonstrating every pattern needed to go from a monolith to independently deployable, event-driven services. It ships with a **13-part blog series**, **seven Grafana dashboards**, and **10 AI-powered MCP tools** — ready for demos, PoCs, enablement, and blog content.

The "What" (Everyone Says This)

"Use Hazelcast for in-memory data"

"Hazelcast is fast and scalable"

"Put Hazelcast in your microservices"

"Hazelcast supports event streaming"

The "How" (This Framework Shows It)

- Event sourcing with IMap, Event Journal, and Jet pipelines
- 100K+ ops/sec views at <1ms P99 via materialized views
- Distributed sagas coordinated through ITopic pub/sub
- Write-behind persistence, circuit breakers, dead letter queues

Hazelcast Features in Action

IMap
Event store, views, saga state, outbox, DLQ

Jet Pipeline
Real-time stream processing

ITopic
Cross-service event bus

Event Journal
Change data capture

Entry Processor
Atomic in-place updates

MapStore
Write-behind to PostgreSQL

Near Cache
Local query acceleration

FlakeIdGenerator
Distributed sequence IDs

VectorCollection
AI similarity search (EE)

Live Dashboards — Demo-Ready Out of the Box

Business Overview — Orders, Revenue, Customers

Saga Coordination — Distributed Transactions

Proven Performance & AI-Ready

Traditional Hazelcast Strengths — Demonstrated End-to-End

High Throughput

Materialized views deliver **100,000+ operations/sec** with sub-millisecond latency. Customers see these numbers live in the Grafana dashboard, not in a slide deck.

Low Latency at Scale

End-to-end event processing (HTTP → pipeline → view update → publish) completes in **<13ms at P50**. The Jet pipeline runs entirely in-memory with no external message broker.

Resilience & Reliability

Circuit breakers, transactional outbox, dead letter queues, and idempotency guards demonstrate that Hazelcast applications handle **partial failure gracefully** — not just when everything works.

Distributed Coordination

Two saga patterns (choreographed + orchestrated) show Hazelcast ITopic and IMap coordinating **multi-service transactions** with automatic compensation — replacing the need for XA or external workflow engines.

The AI story builds on the performance story. Hazelcast's sub-millisecond latency and in-memory processing aren't just fast for traditional workloads — they're what AI applications need for real-time inference, similarity search, and agentic tool execution. This framework shows the transition path.

New: AI-Ready Capabilities

Vector Similarity Search

Product recommendations via **VectorCollection with HNSW indexing** (Enterprise) or IMap-based cosine similarity (Community). One API endpoint, automatic edition detection, zero code changes between editions.

AI Agent Integration (MCP)

10 MCP tools let AI assistants (Claude, GPT, etc.) query views, submit events, inspect sagas, manage DLQ entries, and run demo scenarios. Supports API key RBAC with three roles.

Content & Enablement Assets

13
Blog Posts
From event sourcing fundamentals through performance engineering. Each maps to a code milestone.

7
Grafana Dashboards
Business metrics, system health, event flow, sagas, views, persistence, performance.

5 min
Time to Demo
Clone, build, docker compose up. Sample data, demo scripts, and dashboards auto-provision.

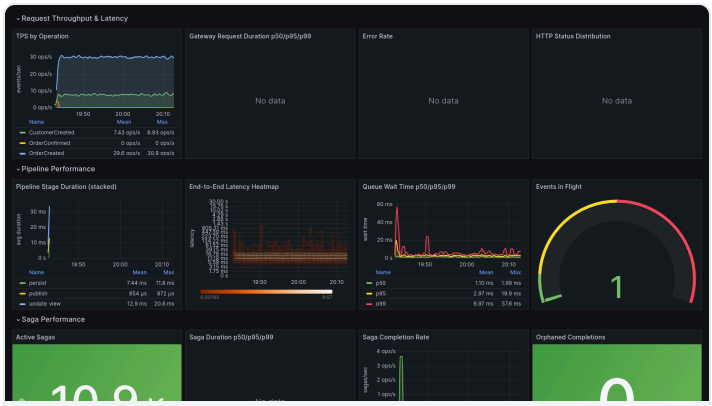
Deployment flexibility: Runs on a developer laptop via Docker Compose, deploys to AWS EKS with provided Helm charts and Terraform, or targets GCP GKE / Azure AKS. Suitable for live demos, trade show booths, and customer PoCs.

Live Dashboards — See It Running

Every claim on the previous page has a dashboard to back it up. Seven pre-provisioned Grafana dashboards auto-deploy with the application — no manual setup, no configuration. Walk a prospect from business outcomes to technical implementation in a single demo session.

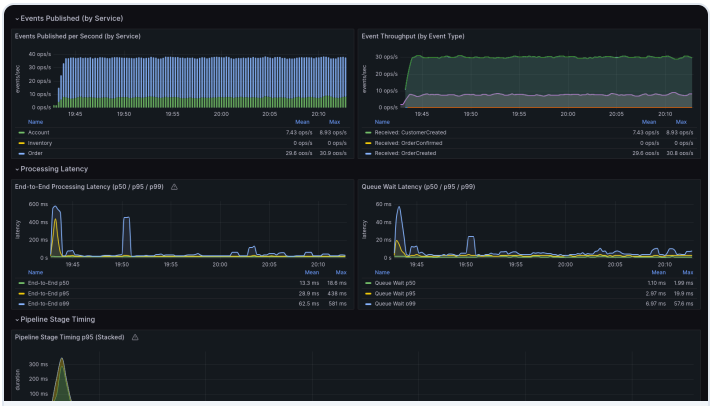
System Overview

Service health, JVM metrics, throughput, latency percentiles, error rates



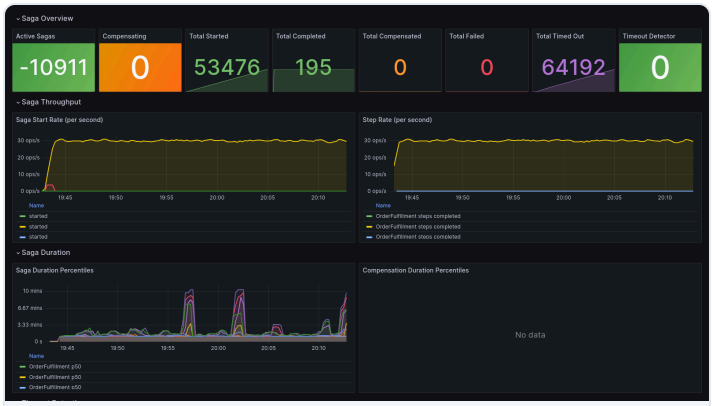
Performance Testing

Live TPS, pipeline throughput, saga completion rates, resource utilization



Event Flow

Event throughput by type, latency percentiles, pipeline stage timing

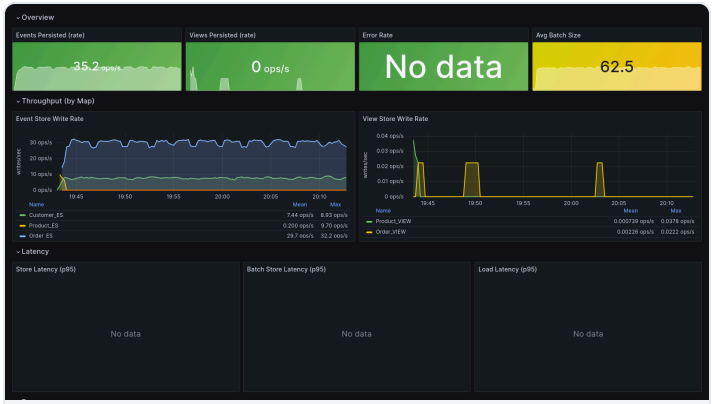


Saga Coordination

Saga outcomes, compensation tracking, step duration, timeout detection

Materialized Views

View rebuild latency, cache hit rates, entry counts, near cache performance



Persistence Layer

Write-behind queue depth, batch sizes, PostgreSQL latency, eviction metrics

How Product Marketing Can Use This

Customer Demos & PoCs

Run the full four-service eCommerce demo live. Show real throughput numbers, trigger saga failures, watch circuit breakers trip — all visible in Grafana. Customize for specific verticals by extending the domain.

Blog & Content Marketing

13 ready-to-publish blog posts covering event sourcing, stream processing, distributed transactions, resilience, persistence, observability, AI integration, and performance engineering. Each post maps to working code.

Trade Shows & Booth Demos

Configurable for long-running demos (8+ hours at low TPS). Dashboards show live business metrics (orders, revenue, customers) alongside technical telemetry. AI assistant interaction via MCP makes an engaging live demo.

SA & Partner Enablement

Architectural decision records (ADRs), guides for every pattern, and a modular codebase that SAs can fork and extend. The blog series doubles as a self-paced technical curriculum.

Technology Stack

Component	Technology	Component	Technology
Runtime	Java 17+	AI / ML	Spring AI MCP 1.0, VectorCollection
Framework	Spring Boot 3.2	Metrics	Micrometer / Prometheus
Data Grid	Hazelcast 5.6	Dashboards	Grafana 10.3
Streaming	Hazelcast Jet	Tracing	Jaeger (OTLP)
Database	PostgreSQL 16+	Containers	Docker / K8s Helm

Accompanying Blog Series (13 Parts): From event sourcing fundamentals through saga patterns, vector search, AI integration, resilience, persistence, observability, and performance engineering. Each post walks through architectural decisions with working code — ready for theyawns.com, the Hazelcast corporate blog, or partner publications.

Ready to Demo

Five minutes from clone to running demo. Docker Compose brings up all four services, a 3-node Hazelcast cluster, PostgreSQL, Prometheus, Grafana, and Jaeger. Sample data and demo scripts are included. The MCP server auto-registers with Claude Code.

Repository: Available upon request
Blog: theyawns.com
Quick Start: ./scripts/docker/start.sh
Contact: mike.yawn@hazelcast.com